TLAM USER GUIDE

This document is the user guide for the A**T**R **L**anding **A**nalysis **M**odule **TLAM** version B2/O9 OCT.2025. This TLAM version meets requirements of IATA / *SCAP Interface Specification - Landing* / 17th Edition September 2017.

B2: SCAP landing interface version number

O9: FOS module1 computation program version number

OCT.2025: TLAM release date

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1. Introduction

All airlines must provide performance data to enable their crews to operate aircraft safely, economically and in accordance with their applicable certification rules and operating practices. Basic data has in the past been provided by the aircraft manufacturer in various manuals (AFM, Operations Manual, Performance Engineer's Manual) in hard copy form. Progressively this has been supplemented by data provided digitally to facilitate data entry into programs developed either by the airlines or by aircraft manufacturers.

Most airlines use programs supplied by the aircraft manufacturers for each aircraft type, and interface these programs with their own computer operating environment, airfield database, and data output format requirements.

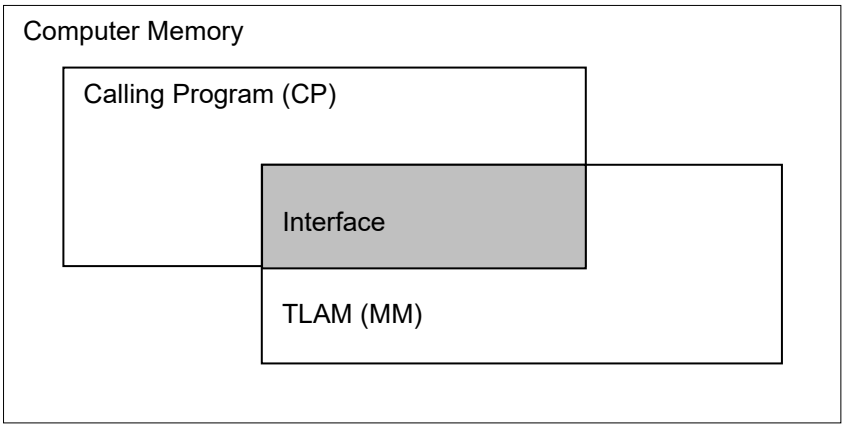
In order to reduce the development effort for the airlines (interfacing with different aircraft manufacturers' programs) and manufacturers (meeting individual airline requirements for customer specific features) there are clear advantages in standardizing as much as possible the input/output features of these programs, and also data formats.

Having recognized this potential benefit, IATA, ATA and several manufacturers set up a Standard Computerized Airplane Performance (SCAP) Working Group. The SCAP Working Group try to achieve maximum possible standardization of the landing performance program interface. The result of that effort is a Specification which defines the interfaces requirements for aircraft manufacturer supplied landing analysis software for implementation in the airline user environment.

The SCAP landing specification is applicable to civil, transport category, aircrafts and are published as joint IATA/ATA documents. Copies of the specification may be ordered at IATA.

2. General

The IATA specification describes the interface between a landing computation Manufacturer Module (MM) and an airline's calling program (CP). This interface is a part of the storage area of the computer memory that is shared by the MM and the CP as shown in the following figure.



The MM is provided by the aircraft manufacturer. It contains logic and data which allow to compute landing performance for a single point calculation, i.e. one temperature, one pressure altitude, one wind and one set of runway data. The MM is to be used as a single subroutine. The MM, called **TLAM** (A**T**R **L**anding **A**nalysis **M**odule), is available for all ATR42 and ATR72 models. The TLAM meets the requirements of the IATA landing specification. TLAM is an input/output interface which is plugged on top of the existing basic FOS Module 1 computation program.

TLAM meets the requirements of the IATA specification as regards the single point computation process philosophy and the interfaced input/output parameters. It contains no optional features conflicting with the specification but may contain some minor deviations which are clearly identified in this user guide.

The CP is implemented by the airline and it contains the procedure that suit the airline's needs. The CP is the main program.

Airline's CP and ATR's MM TLAM can transfer data to and from through the interface common area. CP and MM must be written in FORTRAN 77 programming language and must be compiled and linked together prior to execution.

The CP and the MM must contain exactly the same definition of COMMON BLOCKS and CHARACTER statements as described in the Input interface (Chapter 3) and the Output interface (Chapter 4) which means same COMMON labels, same parameter positions within the COMMON, same array dimensions and character lengths. Input and output data can be created using ASCII-based or XML standards.

The CP initialises the input parameters with actual values, or 9×10^{20} (for unspecified numerical parameters) or blanks (for unspecified alphanumerical parameters). Then, the CP calls the TLAM subroutine which process the input, reads the performance database, computes the case and fills the output interface with the output results or default values. The default values will be equal to 9×10^{20} for numerical parameters or blank for alphanumerical parameters.
Finally, the CP processes the output and ends.

3. Input data

The landing input interface consists of two FORTRAN COMMON BLOCKS: a numerical COMMON BLOCK labelled LINPN and an alphanumerical COMMON BLOCK labelled LINPA.

3.1. Numerical Common Block LINPN

The numerical COMMON BLOCK labelled LINPN contains 5 one-dimensional arrays of single precision REAL, each identified by a 4 character name. FORTRAN 77 conventions imply REAL*4 format for all data in this COMMON BLOCK.
The 4 character names of the arrays in LINPN are the following:

- POPTL Program options
- CONFL Aircraft configuration
- XMETL Meteorological conditions
- RWYDL Runway data
- UNITL Units

Both CP and MM must contain the same following declaration statements:

```
COMMON /LINPN/POPTL(200),CONFL(200),XMETL(200),RWYDL(200),UNITL(200)
```

3.1.1. POPTL – Program options

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Units**	**Comments**
POPTL(1)	CALCULATION OPTION. Allows the user to produce different levels of output.	0. = Full output 1. = Short output 2. = N/A	NONE	
POPTL(2)	SURFACE CONDITION/TYPE OF CONTAMINATION. Different types of runway contamination. Only if POPTL(36) = 0	0. = Dry 1. = Wet 2. = Standing water 3. = Slush 4. = Compacted snow 5. = N/A 6. = N/A 7. = Ice 8. = N/A 9. = N/A 10.= N/A 11.= N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. For valid values 2., 3., 4., and 6., when certified data is computed the result should not be greater than the wet result (1. value).
POPTL(3)	DEPTH OF CONTAMINATION. Applicable only if POPTL(2) = 2 or 3	.25 = 1/4 inch .50 = 1/2 inch	UNITL(5)	

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPTL(4)	SELECTED MU OR BRAKING ACTION VALUE	Greater than 0. & less than 1. = Input Airplane Braking MU value 0.= Nil 1.= 1-Poor 2.= 2-Medium to Poor 3.= 3-Medium 4.= 4-Medium to Good 5.= 5-Good 6.= 6-Dry 7.= 6-Dry Unpaved 8.= 6-Dry Grass 9.= 5-Good Gravel 10.= 3-Dry snow (11-25mm) 11.= 3-Dry snow (26-50mm) 12.= 3-Wet snow (6-10mm) 13.= 3-Wet snow (11-20mm) 14.= 2-Water/Slush (6-12mm)	NONE	
POPTL(5)	TRACE FLAG LEVEL.	0. = No trace 1. = Trace	NONE	
POPTL(6)	LOGICAL UNIT NUMBER FOR TRACE OUTPUT Debugging results and ERROR messages are directed to this file.	Any Logical Unit Number valid for output in user's installation.	NONE	A Logical Unit Number is an integer. Notice that POPTL(6) is a REAL type variable.
POPTL(7)	LOGICAL INPUT UNIT NUMBER FOR DATABASE. Needed as the database is read from an external file.	Logical number of the file containing the Database.	NONE	
POPTL(8)	SELECTED MAXIMUM COMPUTATIONAL WEIGHT FLAG.	0. = AFM performance charts limit 1. = Selected value. [Value to be set in POPTL(9)].	NONE	User can limit computed weight by a "structural" limiting weight or by any other weight. Note that output flight path and speeds are computed for the resulting limited landing weight CLIMIL(1).
POPTL(9)	SELECTED MAXIMUM COMPUTATIONAL WEIGHT VALUE. Only when POPTL(8) = 1.		UNITL(8)	
POPTL(10)	LANDING WEIGHT VALUE FOR CRITICAL TEMPERATURE	N/A	UNITL(8)	
POPTL(11)	DERATED GO-AROUND THRUST LEVEL.	N/A		
POPTL(12)	FORWARD CENTER OF GRAVITY POSITION LIMIT FLAG.	0. = Standard (basic) 1. = N/A 2. = N/A	NONE	
POPTL(13)	THRESHOLD SPEED FLAG	0. = Normal (Vref) 1. = N/A 2. = N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPTL(14)	THRESHOLD SPEED INCREMENT OR FACTOR. Only if POPTL(13) = 1 or 2.	N/A		
POPTL(15)	VISIBILITY	0. = Normal 1. = N/A	NONE	
POPTL(16)	AIRPORT SELECTION	0. = Destination 1. = N/A	NONE	
POPTL(17)	APPROACH CLIMB GROSS GRADIENT FLAG.	0. = Normal 1. = Selected gradient value	NONE	
POPTL(18)	APPROACH CLIMB GROSS GRADIENT VALUE. If POPTL(17) = 1.		%	AFM chart boundary check inside computation module.
POPTL(19)	APPROACH CLIMB SPEED FLAG	0. = Normal 1. = N/A 2. = K * Vs	NONE	
POPTL(20)	APPROACH CLIMB SPEED DECREMENT / INCREMENT OR FACTOR. Only if POPTL(19) = 1 or 2.			
POPTL(21)	LANDING CLIMB GROSS GRADIENT FLAG	0. = Normal 1. = Selected gradient value	NONE	
POPTL(22)	LANDING CLIMB GROSS GRADIENT VALUE. If POPTL(21) = 1.		%	AFM chart boundary check inside computation module.
POPTL(23)	LANDING CLIMB SPEED FLAG	0. = Normal 1. = N/A 2. = N/A	NONE	
POPTL(24)	LANDING CLIMB SPEED DECREMENT / INCREMENT OR FACTOR. Only if POPTL(23) = 1 or 2.	N/A		
POPTL(25)	ENGINE STATUS FOR CLIMB LIMIT	0. = Normal 1. = N/A 2. = N/A 3. = N/A	NONE	
POPTL(26)	APPROACH CATEGORY CHOICE	0. = Normal approach 1. = N/A		
POPTL(27)	BLANK SPACE AVAILABLE			
POPTL(28)	DRAG INDEX	N/A		
POPTL(29)	TIME HISTORY FLAG	0. = No time History other = N/A	NONE	
POPTL(30)	LOGICAL UNIT NUMBER FOR TIME HISTORY OUTPUT	N/A	NONE	

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
POPTL(31)	AIRPLANE CENTER OF GRAVITY	N/A	NONE	
POPTL(32)	CALCULATION TYPE	N/A		
POPTL(33)	THRESHOLD SCREEN HEIGHT	N/A	UNITL(4)	
POPTL(34)	LANDING PROCEDURE	0. = Normal braking 1. = Delayed braking	NONE	
POPTL(35)	BLANK SPACE AVAILABLE			
POPTL(36)	LANDING DISTANCE CALCULATION MODE	0. = Dispatch/RLD 1. = Operational/OpLD	NONE	
POPTL(37)	BLANK SPACES AVAILABLE			
POPTL(38)	PERFORM WET CHECK	0. = Yes (Default) 1. = No	NONE	Only if POPTL(36) = 0.
POPTL(39) thru POPTL(200)	BLANK SPACES AVAILABLE			

3.1.2. CONFL – Aircraft configuration

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONFL(1)	SLAT/FLAP CHOICE.	0. = N/A 1. = Selected configuration 2. = N/A 3. = N/A 4. = Selected configuration (alphanumerical value). Refer to FDCFAPP and FDCFLD in common LINPA.	NONE	If CONFL(1) = 1. let FDCFGA and FDCFGL blank. If (CONFL(1) = 4. let CONFL(2), CONFL(3), CONFL(4) and CONFL(5) unassigned.
CONFL(2)	APPROACH SLAT POSITION.	0.	Degrees	No slats on ATR
CONFL(3)	APPROACH FLAP POSITION.	Refer to AFM for valid flap positions	Degrees	E.G.: 15.
CONFL(4)	LANDING SLAT POSITION.	0.	Degrees	No slats on ATR
CONFL(5)	LANDING FLAP POSITION.	Refer to AFM for valid flap positions	Degrees	E.G.: 35.
CONFL(6)	AIR CONDITIONING.	0. = N/A 1. = N/A 2. = Off 3. = N/A 4. = N/A	NONE	For CAT 2 landing (CONFL(30)=2.) results are given with air conditioning on.
CONFL(7)	ANTI-ICING.	0. = Off 1. = N/A 2. = N/A 3. = N/A 4. = N/A 5. = N/A 6. = Atmospheric icing conditions.	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(8)	GROUND SPOILERS.	0. = N/A 1. = N/A 2. = N/A 3. = N/A 4. = N/A	NONE	
CONFL(9)	ANTISKID.	0. = On 1. = N/A 2. = Totally off	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(10)	THRUST REVERSE CREDIT.	0. = No reverse thrust credit 1. = Credit for all thrust reversers 2. = N/A 3. = N/A 4. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(11)	REVERSE THRUST LEVEL SELECTION. If CONFL(10) not equal to 0.	0. = Normal 1. = N/A 2. = N/A 3. = N/A 4. = N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONFL(12)	LANDING GEAR POSITION FOR APPROACH, LANDING AND GO-AROUND.	0. = Normal 1. = Extended 2. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. Value 1. cannot be combined with atmospheric icing condition CONFL(7) = 6.
CONFL(13)	MAIN FUSELAGE LANDING GEAR POSITION FOR LANDING.	N/A		
CONFL(14)	APU.	N/A	NONE	
CONFL(15)	POWER MANAGEMENT SYSTEM.	0. = Normal 1. = N/A 2. = N/A 3. = N/A 4. = one torque indicator inoperative 5.= N/A 6. = N/A 7. = N/A 8. = N/A 9. = one ACW generator inoperative	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(16)	AUTO BRAKES CHOICE.	N/A		
CONFL(17)	AUTO BRAKES SETTING.	N/A		
CONFL(18)	TIRE PRESSURE.	0. = Normal 1. = N/A		
CONFL(19)	TIRE PROFILE.	0. = Normal 1. = N/A		
CONFL(20)	TIRE SPEED.	N/A		
CONFL(21)	BRAKE ENERGY CHOICE.	0. = Normal 1. = Selected value		
CONFL(22)	BRAKE ENERGY VALUE.	Refer to AFM for valid values	MJ	
CONFL(23)	SPEED BRAKES.	N/A	NONE	
CONFL(24)	BRAKES CONFIGURATION.	0. = All brakes operating 1. = one Brake inoperative 2. = N/A 3. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(25)	TIRE / WHEEL CONFIGURATION.	N/A		
CONFL(26)	HYDRAULIC PUMP INOPERATIVE.	N/A		
CONFL(27)	ONE ENGINE OUT FERRY FLIGHT.	N/A		
CONFL(28)	SPARE ENGINE CARRIAGE.	N/A		
CONFL(29)	RECONTOURNED TAIL LEADING EDGE.	N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
CONFL(30)	LANDING CATEGORY CHOICE.	0. = N/A 1. = Manual landing 2. = Approach category 2 3. = Steep approach 4. = N/A	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface.
CONFL(31)	HEIGHT OF ILS REFERENCE DATUM. Applicable only if CONFL(30) = 0, 3 or 4.	N/A		
CONFL(32)	GLIDE PATH ANGLE. Applicable only if CONFL(30) = 0, 3 or 4.	+ (Positive)		Only if CONFL(30) = 3.
CONFL(33)	VORTEX DISSIPATORS.	N/A		
CONFL(34)	RESIDUAL ICE ACCRETION.	N/A		
CONFL(35)	BRAKE COOLING FANS.	N/A		
CONFL(36)	ROLL SPOILERS.	N/A		
CONFL(37)	SPEEDBRAKES.	N/A		
CONFL(38)	ONE ENGINE INOP. LANDING FLAP.	N/A		
CONFL(39)	AUTOLAND CONTAMINATED DISTANCE FACTOR.	N/A		
CONFL(40)	TACHYMETER FAILURE.	N/A		
CONFL(41)	FLAPS FAULT / LOCKED	0. = No 1. = Retracted	NONE	Provided aircraft database contains relevant tables. Check which conditions are available in FOS Module 1 interface. Value 1. cannot be combined with atmospheric icing condition CONFL(7) = 6.
CONFL(42)	SLATS FAULT / LOCKED	N/A		
CONFL(43)	PROPELLER RPM FOR NOISE ABATEMENT FLAG	N/A		
CONFL(44)	PROPELLER RPM VALUE	N/A		
CONFL(45)	BOOST	0. = File default 1. = Boost off 2. = Boost on 3. = Super boost on	NONE	Option 1. to 3. only valid with multi boost file 0. is equivalent to boost off on dual boost file
CONFL(46) thru CONFL(200)	BLANK SPACES AVAILABLE			

3.1.3. XMETL – Meteorological conditions

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Units**	**Comments**
XMETL(1)	TEMPERATURE. Outside Air Temperature (OAT).		UNITL(1)	
XMETL(2)	WIND COMPONENT. Unfactored wind component along runway.	+ (Positive) = Headwind - (Negative) = Tailwind	UNITL(9)	Data calculation is based on wind component measured at 10m.
XMETL(3)	NOT ASSIGNED	N/A		
XMETL(4)	ALTITUDE FLAG. User choice to measure airport altitude	0. = Airport pressure altitude 1. = QNH 2. = N/A	NONE	
XMETL(5)	ALTITUDE VALUE. Altitude for which computation will be performed, expressed in terms of XMETL(4).		UNITL(6)	Units must be in accordance with choice in XMETL(4).
XMETL(6)	GUST COMPONENT. Unfactored gust component along the runway. (+ headwind gust, - tailwind gust)	N/A	UNITL(9)	This effect is separate from the steady wind component in XMETL(2).
XMETL(7) thru XMETL(200)	BLANK SPACES AVAILABLE			

3.1.4. RWYDL – Runway data

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
RWYDL(1)	ELEVATION. Airport geometrical elevation with respect to Sea Level. (Airport Reference Elevation)		UNITL(4)	
RWYDL(2)	LANDING DISTANCE AVAILABLE (LDA).		UNITL(3)	
RWYDL(3)	SLOPE FOR LDA.	+ (Positive) = Up-hill - (Negative) = Down-hill	%	
RWYDL(4)	RUNWAY SURFACE.	0. = Normal 1. = N/A 2. = N/A 3. = Dry unpaved with CBR lower than 15 4. = Dry unpaved with CBR equal or greater than 15 5. = Grass runway with CBR lower than 15 6. = Grass runway with CBR equal or greater than 15 7. = N/A	NONE	Provided aircraft database contains relevant tables. Check which surfaces are available in FOS Module 1 interface.
RWYDL(5)	LDA MULTIPLICATION FACTOR (K). This parameter allows user to multiply RWYDL(2) length by a correction factor.	$0. \leq K \leq 1.$	NONE	LDA for computation = K*LDA + A
RWYDL(6)	LDA INCREMENT VALUE (A). This parameter allows user to apply an increment (decrement if negative) to the actual LDA.		UNITL(3)	
RWYDL(7)	HORIZONTAL DISTANCE FROM 50 FEET ABOVE THRESHOLD TO TOUCHDOWN FLAG.	N/A		
RWYDL(8)	HORIZONTAL DISTANE FROM 50 FEET ABOVE THREHOLD TO TOUCHDOWN VALUE. Only if RWYDL(7) = 1.	N/A	UNITL(3)	
RWYDL(9)	OPERATIONAL DISTANCE FACTOR	Equal or greater than 1. and less than 2. Default is 1.		Applicable to OpLD
RWYDL(10)	OPERATIONAL DISTANCE INCREMENT	+ (Positive) Default is 0.		Applicable to OpLD
RWYDL(11)	CBR (California Bearing Ratio)	N/A		

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
RWYDL(12)	OPERATIONAL LANDING DISTANCE (OpLD) FACTOR/INCREMENT OPTION	0. = apply both (default) 1. = use maximum of factored or incremented distance	NONE	Defines how RWYDL(9) and RWYDL(10) are applied. K*calc. distance + A for RWYDL(12) = 0.
RWYDL(13) thru RWYDL(200)	BLANK SPACES AVAILABLE			

3.1.5. UNITL – Units

Array Position	Parameter	Valid Values (N/A means Not Available)	Units	Comments
UNITL(1)	TEMPERATURE.	0. = Celsius 1. = Fahrenheit 2. = Deviation from ISA - Celsius 3. = Deviation from ISA - Fahrenheit		
UNITL(2)	TIME.	N/A		
UNITL(3)	LENGTH AND HORIZONTAL DISTANCES.	0. = Meters 1. = Feet		
UNITL(4)	HEIGHT, ELEVATION, PRESSURE ALTITUDE.	0. = Meters 1. = Feet		Elevation and Pressure Altitude for Airport and Runway.
UNITL(5)	DEPTH OF CONTAMINATION.	0. = N/A 1. = Inches		
UNITL(6)	PRESSURE (QFE,QNH).	0. = HectoPascal 1. = Inches Hg.		(mb)
UNITL(7)	TIRE PRESSURE.	N/A		
UNITL(8)	WEIGHT.	0. = Kilograms 1. = Pounds 2. = N/A		
UNITL(9)	SPEED (GROUND AND AIR).	0. = N/A 1. = N/A 2. = Knots 3. = N/A		
UNITL(10) thru UNITL(200)	BLANK SPACES AVAILABLE			

3.2. Alphanumerical Common Block LINPA

The alphanumerical COMMON BLOCK labelled LINPA contains 14 parameters and one array of three elements for comments. Every one of these parameters has an independent number of characters and has to be left justified with blanks filled.

Both CP and MM must contain the same following declaration statements:

```
      CHARACTER*20 ACTYPL,APTIDL,BRAKEL,CERTIL,DATEL,DBOPTL,ENGTYL,
     +             FDCFGA,FDCFGL,RATINL,REVNBL,RWYNBL
      CHARACTER*255 AMDBIL,MISCLL
      CHARACTER*132 COMENL(20)
      COMMON /LINPA/ACTYPL,ENGTYL,RATINL,BRAKEL,CERTIL,REVNBL,DATEL,
     +       DBOPTL,AMDBIL,APTIDL,RWYNBL,COMENL,MISCLL,FDCFGA,FDCFGL
```

Array Position	**Parameter**	**Valid Values** (N/A means Not Available)	**Format**	**Comments**
ACTYPL	AIRCRAFT TYPE.	N/A	CHARACTER*20	SEE ACMDBI array
ENGTYL	ENGINE TYPE.	N/A	CHARACTER*20	
RATINL	RATING LEVEL.	N/A	CHARACTER*20	
BRAKEL	BRAKES TYPE.	N/A	CHARACTER*20	
CERTIL	CERTIFICATION BASIS.	Blank for mono regulation file For multi regulation file: JARA for JAR regulation FAR for FAR regulation	CHARACTER*20	Provided aircraft database contains relevant tables. Check which regulations are available in FOS Module 1 interface.
REVNBL	REVISION NUMBER.	N/A	CHARACTER*20	SEE ACMDBI array
DATEL	REVISION DATE.	N/A	CHARACTER*20	
DBOPTL	DATABASE INPUT MODIFICATION OPTIONS.	N/A	CHARACTER*20	
AMDBIL	MANUFACTURER DATABASE IDENTIFICATION CODE. Set of characters, manufacturer defined, that identifies a specific database.	Performance data file name.	CHARACTER*255	Enter here the logical name corresponding to the specified aircraft.
APTIDL	AIRPORT IDENTIFICATION.		CHARACTER*20	
RWYNBL	RUNWAY IDENTIFICATION.		CHARACTER*20	
COMENL(1) thru COMENL(20)	COMMENTS		CHARACTER*132	Allows writing here engine failure procedure or other comments.
MISCLL	MISCELANEOUS. Provision for future input data.	N/A	CHARACTER*255	
FDCFGA	SLAT/FLAP CONFIGURATION FOR APPROACH		CHARACTER*20	E.G.: S00F15.
FDCFGL	SLAT/FLAP CONFIGURATION FOR LANDING		CHARACTER*20	E.G.: S00F15.

4. Output data

The IATA specification do not request manufacturers to provide all specified output parameters but as many parameters as feasible. Output parameters which are not available from the TLAM are identified by the code N/A in the COMMENTS column and are set to 9 x 10^{20}. Output parameters exceeding the limits are set to 9999999 and a message is displayed. The units described in chapter 3.1.5 apply to input parameters as well as to the expected output parameters.

The landing output interface consists of two COMMON BLOCKS : a numerical COMMON BLOCK labelled LOUTN and an alphanumerical COMMON BLOCK labelled LOUTA.

4.1. Numerical Common Block LOUTN

The numerical COMMON BLOCK labelled LOUTN contains one-dimensional array of single precision REAL, each identified by a 6 character name. FORTRAN 77 conventions imply REAL*4 format for all data in this COMMON BLOCK.
The 6 character name of the array in LOUTN is the following:

- CLIMIL Limiting condition

Both CP and MM must contain the same following declaration statement:

COMMON /LOUTN/CLIMIL

4.1.1. CLIMIL – Limiting condition

Array Position	**Parameter**	**Units**	**Comments** (N/A means Not Available)
CLIMIL(1)	THE MAXIMUM LANDING WEIGHT.	UNITL(8)	
CLIMIL(2)	THE LANDING SLAT POSITION.	NONE	N/A
CLIMIL(3)	THE LANDING FLAP POSITION.	NONE	N/A
CLIMIL(4)	THE APPROACH SLAT POSITION.	NONE	N/A
CLIMIL(5)	THE APPROACH FLAP POSITION.	NONE	N/A
CLIMIL(6)	THE LIMITATION CODE. The limitation code values are: 0 =CLIMIL(1) is limited by WET runway performance 1 = The selected maximum computational weight according to POPTL(9) 2 = The available landing distance weight. 3 = The approach climb limited weight. 4 = The landing climb limited weight. 5 = The tire speed limited weight. 6 = The brake energy limited weight.	NONE	Corresponds to FOS module 1 code: - 0 - 1 - 2 - 3 - 4 - N/U - N/U
CLIMIL(7)	THE WEIGHT LIMITED BY AVAILABLE LANDING DISTANCE.	UNITL(8)	
CLIMIL(8)	THE WEIGHT LIMITED BY THE APPROACH CLIMB CONDITION.	UNITL(8)	
CLIMIL(9)	THE WEIGHT LIMITED BY THE LANDING CLIMB CONDITION.	UNITL(8)	
CLIMIL(10)	THE WEIGHT LIMITED BY TIRE SPEED.	UNITL(8)	N/A
CLIMIL(11)	THE WEIGHT LIMITED BY BRAKE ENERGY.	UNITL(8)	
CLIMIL(12)	THE LANDING DISTANCE ASSOCIATED WITH THE MAXIMUM LANDING WEIGHT CLIMIL(1).	UNITL(3)	
CLIMIL(13)	THE REQUIRED RUNWAY LENGTH ASSOCIATED WITH THE MAXIMUM LANDING WEIGHT CLIMIL(1).	UNITL(3)	
CLIMIL(14)	THE APPROACH CLIMB GRADIENT ASSOCIATED WITH THE MAXIMUM LANDING WEIGHT CLIMIL(1)	%	

Array Position	Parameter	Units	Comments (N/A means Not Available)
CLIMIL(15)	THE LANDING CLIMB GRADIENT ASSOCIATED WITH THE MAXIMUM LANDING WEIGHT CLIMIL(1)	%	
CLIMIL(16)	THE MAXIMUM QUICK TURNAROUND WEIGHT.	UNITL(8)	N/A
CLIMIL(17)	THE MAXIMUM QUICK TURNAROUND TIME.	UNITL(2)	N/A
CLIMIL(18)	THE CRITICAL TEMPERATURE ACCORDING TO POPTL(10).	UNITL(1)	N/A
CLIMIL(19)	THE CRITICAL TEMPERATURE ACCORDING TO CLIMIL(1).	UNITL(1)	N/A
CLIMIL(20)	MINIMUM FLAT RATING TEMPERATURE.	UNITL(1)	N/A
CLIMIL(21)	MAXIMUM FLAT RATING TEMPERATURE.	UNITL(1)	N/A
CLIMIL(22)	THE LANDING THRESHOLD SPEED ACCORDING TO CLIMIL(1).	UNITL(9)	N/A
CLIMIL(23)	THE APPROACH CLIMB SPEED ACCORDING TO CLIMIL(1).	UNITL(9)	N/A
CLIMIL(24)	THE LANDING CLIMB SPEED ACCORDING TO CLIMIL(1).	UNITL(9)	N/A
CLIMIL(25)	THE APPROACH MINIMUM CONTROL SPEED (Vmcl) WITH ONE ENGINE INOPERATIVE.	UNITL(9)	N/A
CLIMIL(26)	THE APPROACH MINIMUM CONTROL SPEED (Vmcl) WITH TWO ENGINES INOPERATIVE.	UNITL(9)	N/A
CLIMIL(27)	HORIZONTAL DISTANCE FROM 50 FEET ABOVE THRESHOLD TO TOUCHDOWN (UNFACTORED).	UNITL(3)	N/A
CLIMIL(28)	CRITICAL TAILWIND	UNITL(9)	N/A
CLIMIL(29)	CRITICAL TEMPERATURE (WAT/ Climb Limit)	UNITL(1)	N/A
CLIMIL(30)	Unassigned		N/A
CLIMIL(31)	Unassigned		N/A
CLIMIL(32)	Unassigned		N/A
CLIMIL(33)	Unassigned		N/A
CLIMIL(34)	BRAKING SPEED	UNITL(9)	N/A
CLIMIL(35)	INCREASE IN LDR FOR ABNORMAL FLAP	%	N/A
CLIMIL(36)	VREF DELTA-WING NOT DEICED CONDITION	UNITL(9)	N/A
CLIMIL(37)	VREF DELTA-ICE REMAINING ON AIRFRAME CONDITION	UNITL(9)	N/A
CLIMIL(38)	CLIMB LIMIT WEIGHT REDUCTION WING NOT DEICED CONDITION	UNITL(8)	N/A
CLIMIL(39)	NORMAL/ABNORMAL LANDING FLAP FLAG	NONE	N/A
CLIMIL(40)	LANDING DISTANCE REQUIRED (LDR)	UNITL(3)	N/A
CLIMIL(41)	REGULATED LANDING WEIGHT	UNITL(8)	N/A
CLIMIL(42)	MAXIMUM OPERATING TAILWIND	UNITL(9)	N/A
CLIMIL(43)	GLIDEPATH MINIMUM LANDING WEIGHT		N/A
CLIMIL(44)	ONE ENGINE INOPERATIVE LANDING FLAP	NONE	N/A
CLIMIL(45)	WEIGHT LIMITED BY THE LOW VISIBILITY ALL ENGINE CLIMB CONDITION Applicable when POPTL(15) = 1.	UNITL(8)	N/A
CLIMIL(46)	WEIGHT LIMITED BY THE LOW VISIBILITY ONE ENGINE INOPERATIVE CLIMB CONDITION Applicable when POPTL(15) = 1.	UNITL(8)	N/A
CLIMIL(47)	ACCUMULATED BRAKE ENERGY. Associated with the Maximum Landing Weight, CLIMIL(1).	NONE	N/A
CLIMIL(48)	ONE ENGINE INOPERATIVE THRESHOLD SPEED. Associated with the Maximum Landing Weight, CLIMIL(1).	UNITL(8)	N/A
CLIMIL(49)	GO AROUND POWER SETTING (Inflight T/O Power)	NONE or %	N/A
CLIMIL(50)	MAXIMUM ENVIRONMENTAL LIMIT TEMPERATURE	UNITL(1)	N/A

Array Position	Parameter	Units	Comments (N/A means Not Available)
CLIMIL(51)	LOW VISIBILITY ALL ENGINE CLIMB GRADIENT, AT CLIMIL(1)	%	N/A
CLIMIL(52)	LOW VISIBILITY ONE ENGINE INOPERATIVE CLIMB GRADIENT, AT CLIMIL(1)	%	N/A
CLIMIL(53)	LOW VISIBILITY ALL ENGINE CLIMB SPEED, AT CLIMIL(1)	UNITL(9)	N/A
CLIMIL(54)	LOW VISIBILITY ONE ENGINE INOPERATIVE CLIMB SPEED, AT CLIMIL(1)	UNITL(9)	N/A
CLIMIL(55) thru CLIMIL(200)	Blank spaces available		

4.2. Alphanumerical Common Block LOUTA

The alphanumerical COMMON BLOCK labelled LOUTA contains one parameter for error code, three parameters of information, one array of ten elements for error messages and one array of three elements for miscellaneous information. Every one of these parameters has an independent number of characters and has to be left justified with blanks filled.

Both CP and MM must contain the same following declaration statements:

```
CHARACTER*20 ERFLAL
CHARACTER*20 MMDATL,MMIDNL
CHARACTER*255 AMDBOL
CHARACTER*80 ERRMSL(20),OUTMIL(20)
COMMON /LOUTA/ERFLAL,ERRMSL,MMIDNL,MMDATL,AMDBOL,OUTMIL
```

Array Position	**Parameter**	**Format**	**Comments**
ERFLAL	Error code flag	CHARACTER*4	A = Successful B = Input error C = Computation error D = TLAM has restricted input value to AFM performances chart limits E = Performance restrictions do not allow computation of valid landing weight or speeds F = Successful. Warning: input value(s) not used for this A/C model. See trace file.
ERRMSL(1) thru ERRMSL(20)	Error messages	CHARACTER*80	See also trace output file (logical unit defined by POPTL(6)) for more details on error and warning messages
MMIDNL	Manufacturer Module Identification	CHARACTER*20	Read here : TLAM
MMDATL	Manufacturer Module Date	CHARACTER*20	
AMDBOL	Aircraft Manufacturer database identification code.	CHARACTER*255	Read here same as in input array AMDBIL
OUTMIL(1) thru OUTMIL(20)	Miscellaneous - Provision for future output data	CHARACTER*80	